

5G-Enabled Haptic Navigation System for the Blind-Deaf

Distributed Spatial Intelligence and Haptic Feedback Framework

Development Team

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1 Objective

The objective of this project is to architect and implement a high-precision, real-time tactile navigation framework specifically engineered for individuals who are both blind and deaf. By transitioning away from conventional visual and auditory interaction methods, we aim to facilitate independent mobility through a highly structured haptic interface.

We focus on solving the hardware bottleneck where small wearables lack the battery life and cooling capacity to handle the heavy math required for precise, real-time navigation. We offload the heavy routing math and real-time path tracking to a dedicated 5G Edge server via 5G. This architecture allows the wearable to stay small and power efficient while still providing high-precision guidance.

2 Summary of Proposal

Modern navigation infrastructure remains fundamentally exclusionary, predicated almost entirely on visual maps and auditory voice prompts. For the deafblind demographic, this creates a total disconnect from current smart city infrastructure, making independent travel hazardous and cognitively exhausting. Our design uses a ‘thin-client’ approach: an ESP32 handles the tactile interface on the wrist, while a 5G Edge server manages the high-intensity navigational processing. This separation keeps the hardware lightweight and responsive.

The architecture relies on a bifurcated wearable system: a passive conductive glove for tactile input and a centralized wrist-mounted transceiver module for processing and haptic actuation. By utilizing 5G-enabled WebSockets/MQTT, the device maintains a persistent, full-duplex connection to a Python-based edge server. This server handles the primary computational load, which involves densifying paths to 1-meter resolution, calculating bearing differentials, and map-matching positional updates. This intelligence is then transmitted back to the user as instantaneous directional vibration triggers, ensuring

a low-latency, responsive guidance experience.

3 Detailed Project Proposal

Our implementation is built on the philosophy of **Distributed Intelligence**. To achieve real-time responsiveness without the bulk of a smartphone-class processor, we have split the project into specialized technological pillars.

At the hardware level, the system is powered by an ESP32 microcontroller utilizing a dual-core architecture. We have strategically partitioned the system firmware to run on Core 0 and Core 1 simultaneously, ensuring that heavy networking tasks never interfere with time-critical user inputs.

3.1 Real-Time OS Architecture (FreeRTOS)

The wrist module operates on FreeRTOS (Real-Time Operating System), which is fundamental for maintaining the safety-critical nature of the device. FreeRTOS allows us to move beyond simple sequential code execution into a preemptive, priority-based multitasking environment. We utilize this to ensure that time-sensitive tasks such as hardware interrupts from the glove or incoming haptic commands from the 5G network are never delayed by background operations. By implementing task notification and binary semaphores, we prevent resource contention, ensuring that data from the GPS module and the WebSocket data stream never overlap or corrupt, providing a seamless user experience.

3.2 Core 0: Input Decoding & Character Mapping Engine

Core 0 is dedicated to user interaction and state logic, managing the sophisticated character-mapping system that allows for data entry without a visual interface.

- **5-Bit Encoding Logic:** Every alphanumeric character is mapped to a unique 5-bit sequence of dots and dashes. As the user presses the glove buttons, Core 0 populates a frame buffer. Once a 5-bit sequence is completed, a decoding algorithm translates that bit-pattern into its corresponding character (A–Z, 0–9, or space).
- **System Buffer Management:** Decoded characters are pushed into a primary destination buffer, which stores the full address string. The system provides real-time tactile confirmation for every completed character, ensuring the user knows their input was registered.
- **Contextual Modal Repurposing:** Core 0 dynamically shifts the button functions based on three operational states:
 1. *Input Mode:* Buttons are used to build the destination string via dot-dash sequences.
 2. *Review Mode:* A fail-safe check to confirm the typed destination before transmission to the server.
 3. *Navigation Mode:* Buttons are repurposed for live commands: requesting a “resay” of the last instruction, pausing or resuming the trip, or triggering a manual reroute.

3.3 Core 1: Positional Sync & Persistent 5G Channel

Core 1 serves as the dedicated communication and synchronization gateway, handling the high-speed data requirements of the 5G network.

- **Positional Retrieval:** Core 1 constantly polls the GPS module for raw NMEA sentences. It utilizes a mutex-locked shared memory architecture to ensure the server always receives the most accurate coordinate data without interrupting the input task on Core 0.
- **Channel Maintenance:** This core initiates and maintains the persistent 5G WebSocket tunnel. It manages the low-level handshake and monitors the server's heartbeat to ensure the connection is active. A kill-switch mechanism forces a socket drop if no ping, pong, or message is received within 10 seconds, preventing silent connection failures.
- **Navigation Data Flow:** During active navigation, Core 1 transmits periodic GPS coordinates to the server every 5 seconds, allowing the edge-based math engine to track the user relative to the 1-meter densified path. It also monitors an internal FreeRTOS message queue for user commands, instantly injecting triggers like "reroute" or "pause" into the WebSocket stream for immediate server-side execution.
- **Auto-Resume Engine:** On connection loss during an active session, the system saves the destination into a recovery buffer, suspends navigation, and arms an auto-resume flag. On reconnection, the destination is automatically replayed to the server without requiring user re-entry.

3.4 The Edge Navigational Server (Python)

The server-side engine is a custom precision routing engine designed to solve the spatial tracking challenges of pedestrian navigation.

- **1-Meter Path Densification:** The Google Maps Directions API provides sparse polyline waypoints. Our server applies the Haversine formula to interpolate the route, generating nodes at exact 1-meter intervals for high-resolution tracking.
- **Bearing Differential Logic:** The engine calculates the angular difference between consecutive path segments. If the bearing change exceeds 22 degrees, a turn action point is flagged and its instruction is dispatched exactly 1 meter before the actual turn node to account for human reaction time.
- **GPS Noise Filtering:** A 4-meter catch radius and a map-matching algorithm snap erratic GPS coordinates back to the densified path, ensuring haptic pulses only fire when the user is truly aligned with the route.

3.5 Haptic Output Layer: Translating Instructions into Touch

The final and most critical stage of the pipeline is the translation of server instructions into physical vibration patterns that a deafblind user can interpret without sight or sound. When the ESP32 receives a message over the WebSocket, the `onWebsocketMessage()` handler on Core 1 immediately decodes the instruction type and triggers the corresponding vibration motor pattern on the wrist module.

Each navigational state is encoded as a distinct tactile signal:

Server Instruction	Haptic Pattern	Meaning
KEEP STRAIGHT	· — · single long pulse	Continue walking forward
TURN_LEFT	· · — two-short, one-long	Turn left at the next junction
TURN_RIGHT	— · · one-long, two-short	Turn right at the next junction
DRIFT	— — — SOS burst	Off-path, stop and reorient
ARRIVED	· · · · · rapid finish buzz	Destination reached

Table 1: Haptic vibration vocabulary: server instruction to wrist motor pattern mapping.

Turn instructions are deliberately sent 1 meter before the actual turn node. This pre-warning window accounts for the user’s natural reaction time at a walking pace of approximately 1.2–1.4 m/s, ensuring the physical cue arrives before the user reaches the decision point rather than at it.

The system also encodes non-navigational events haptically. On connection loss, a rapid SOS-style burst alerts the user that the link has dropped and navigation is suspended. On session termination, a distinct long-short-long pattern signals that the session has ended cleanly, prompting the user to re-enter a destination if needed.

This vocabulary is intentionally minimal. Limiting the pattern set to five distinct signals reduces the cognitive load on the user and shortens the learning curve for new users, a critical consideration for an assistive device that must be operated entirely by touch.

4 Societal & Industrial Impact

Societal Impact: This project represents a significant advancement in digital inclusion. For the deafblind, autonomy is often a forgotten concept. By providing a reliable, private, and discreet navigation system, we are restoring people’s **independence** and **dignity**. It allows access to education, employment, and public infrastructure that were previously unreachable without a full-time human guide.

Industrial Impact: The architecture of offloading sensory processing to the 5G edge has massive industrial potential. Personnel in zero-visibility environments, such as firefighters in smoke-filled buildings or miners in underground tunnels, can utilize this exact haptic-guidance system. By integrating localized beacons and the tactile feedback glove, workers can navigate safely without relying on sight or sound.

5 Scalability & Market Readiness

The Distributed Intelligence architecture of this system provides an inherent advantage in terms of global scalability and immediate deployment readiness.

- **Scalable and Modular Architecture:** We designed the framework to scale seamlessly from our current campus prototype to a full city-wide network. The modularity of the routing engine and firmware means we can expand the service area or add new features on the server side without ever needing to modify the user’s wearable hardware.
- **Deployment Readiness via Live 5G Infrastructure:** Unlike systems validated on simulated or emulated networks, our framework is built and tested using actual 5G lab hardware. This ensures the prototype is immediately deployable and maintains high-fidelity, deterministic performance under real-world 5G conditions. By validating the system in a live lab environment, we have almost bridged the gap between a theoretical prototype and a production-ready assistive tool.

6 Advanced 5G Technology Utilization

The viability of this framework is fundamentally predicated on the unique architectural advantages of 5G. While previous cellular generations focused on throughput, 5G introduces structural features that allow our system to operate as a high-fidelity, safety-critical interface. Our implementation specifically leverages the following 5G pillars:

- **Ultra-Reliable Low-Latency Communication (URLLC):** In tactile navigation, timing is the difference between a safe turn and a hazardous collision. Conventional 4G or public Wi-Fi introduces non-deterministic jitter and latencies that can exceed 100 ms, which is unacceptable for haptic guidance. By utilizing the 5G URLLC standard, we achieve sub-30 ms deterministic latency. This ensures that when the server detects the user is 1 meter from a turn, the haptic trigger packet reaches the wrist module instantly, providing the user with a real-time physical cue that matches their walking pace.
- **Multi-access Edge Computing (MEC) & The 5G Server:** Our “Thin Client” strategy is entirely enabled by MEC. By hosting our custom Python geometric processor at the 5G Edge (the local base station) rather than a centralized cloud, we eliminate the backhaul latency of the open internet. This MEC-resident server acts as a localized brain, performing high-frequency path-densification and map-matching within the same network hop as the user. This creates a “local loop” feel, where the computational power of a high-end server is effectively brought to the user’s wrist without the weight, heat, or battery drain of a local processor.
- **Network Slicing for Mission-Critical Isolation:** Public networks are often congested with high-bandwidth consumer traffic (e.g., video streaming). To prevent this from interfering with life-saving navigation, we utilize 5G Network Slicing. This allows us to create a virtualized, isolated slice of the network specifically for this assistive device. This slice is protected from public congestion, ensuring that the navigation WebSocket stream maintains a guaranteed bit rate and never competes for resources with non-essential data.
- **High-Bandwidth Foundation for Future AI:** While our current system focuses on haptic coordinates, the 5G Enhanced Mobile Broadband (eMBB) layer is the cornerstone of our future roadmap. The massive uplink capacity of 5G will allow us to stream high-definition environmental camera data from the user to the Edge server

for real-time obstacle detection and computer vision, a feat that would be impossible on any other existing wireless standard.